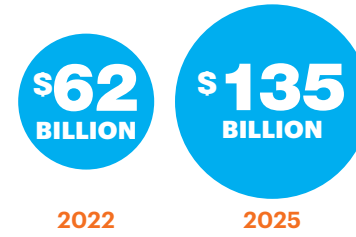
The building blocks of Artificial Intelligence (AI) have been around for decades—since the 1950s, to be precise—and AI tech is something we use daily. Even the ordinary email spam filter is AI too! But an inflection point came in the last decade when three key ingredients converged—faster computing power, tonnes of digital data and advanced algorithms. It pioneered new architecture for an idea as old as AI: neural networks, which were conceived of as a set of parallel computing systems, or nodes, designed to mimic the structure of the human nervous system. Running on powerful algorithms and feeding on terabytes of data, these souped-up neural networks took deep learning to an entirely new level. Enter generative AI which can create all kinds of content—text, images, video—using all that learning material. The most familiar example is ChatGPT, which, when released last November, took the world by storm, kicking off an AI race.

WHY IT IS A GAME CHANGER

If computer vision enabled machines to make sense of images, the latest advances are in Natural Language Processing (NLP)—this is why ChatGPT can converse with humans easily. Chatbots are an immediate use-case for this tech, with conversational AI platforms being deployed to engage with online customers in a wide range of industries: retail, travel, banking, et al. AI is also being built into various devices, such as autonomous driving systems for cars. For now, however, generative AI is the frontier that has captured everybody's imagination—the goal being to boost productivity by taking over mundane jobs.

A GROWING BIZ

The global AI software market is fast expanding



Source: Gartner



"The pace of product development has really accelerated because of generative AI. I never thought we could build products this fast"
– SRIKANTH VELAMAKANNI
Co-founder & Group Chief Executive, Fractal Analytics

THE CHANGE-MAKERS

FRACTAL ANALYTICS

- An early entrant into the AI space in 2000, it started with predicting credit card payment defaults and understanding consumer behaviour during recessions
- It now works with 150 of the Fortune 500 companies

TATA ELXSI

- As a product technology company, applied AI is an integral part of its three core areas: transportation, media & telecom and medical devices
- For instance, on the anvil are advanced driver monitoring systems to keep a tab on the driver's level of alertness and health parameters

INFOSYS

- The Indian IT major is betting on generative AI to drive efficiency and productivity for enterprises
- A new set of services, solutions and platforms, called Topaz, for instance, completed 2,000 processes related to customers at a British bank in real-time instead of a week

WHAT INDIA NEEDS TO MASTER

Generative AI requires deep learning systems known as Large Language Models (LLMs), which are expensive, demanding high computing power and huge volumes of data. "Most of the LLMs available are not necessarily trained on Indian data. So sometimes the context may be missing," says Sangeeta Gupta, senior vice-president & chief of strategy at India's software industry body NASSCOM. An India AI stack, when it comes, could be the base for multiple local applications, in areas such as agriculture and healthcare.

"One in three AI researchers worldwide is an Indian," says Srikanth Velamakanni, co-founder and Group CEO of Fractal Analytics. "So, we have a lot of good talent and we have lots of problems to solve." But building a foundation model for India, he says, requires a mission-scale effort with large investments from both government and technology companies. ■